HTML5 API

Geolocation

```
if (!navigator.geolocation) return;

navigator.geolocation.getCurrentPosition(function(position) {
    //position.coords.latitude
    //position.coords.longitude
}, function(error){
    // errors
});
```

Device Orientation

```
window.addEventListener('deviceorientation', function(event) {
  var a = event.alpha;
  var b = event.beta;
  var g = event.gamma;
}, false);
```

Browser Notification

```
if (window.webkitNotifications.checkPermission() == 0) {
  // you can pass any url as a parameter
  window.webkitNotifications.createNotification(picture,title,text).show();
} else {
  window.webkitNotifications.requestPermission();
}
```

AJAX

WebSockets

```
//Full-Duplex
//bi-directional
//no HTTP overhead

var socket = new WebSocket('ws://app.domain.com');
    socket.onopen = function(event) {
    socket.send('Hello, WebSocket');
};
socket.onmessage = function(event) { alert(event.data); }
socket.onclose = function(event) { alert('closed'); }
```

Offline Application Cache

```
// HTML
<html manifest="cache.appcache">

// cache.appcache (Content-Type: text/cache-manifest)
CACHE MANIFEST
# version 1.0.0
CACHE:
js/logic.js
css/style.css
images/background.png
NETWORK:
*

// JavaScript
window.applicationCache.addEventListener('updateready', function(e) {
    if (window.applicationCache.status == window.applicationCache.UPDATEREADY) {
    window.applicationCache.swapCache();
    if (confirm('A new version of this site is available. Load it?')) {
        window.location.reload();
    }
    }
}, false);
```

Data URIs

```
//HTML
<img src="
iVBORw0KGgoAAAANSUhEUgAAAAoAAAAKCAYAAACNMs+9AAAABGdBTUEAALGP
C/xhBQAAAAlwSFlzAAALEwAACxMBAJqcGAAAAAd0SU1FB9YGARc5KB0XV+IA
AAAddEVYdENvbW1lbnQAQ3JlYXRlZCB3aXRoIFRoZSBHSU1Q72QlbgAAAF1J
REFUGNO9zL0NglAAxPEfdLTs4BZM4DIO4C7OwQg2JoQ9LE1exdlYvBBeZ7jq
ch9//q1uH4TLzw4d6+ErXMMcXuHWxId3KOETnnXXV6MJpcq2MLaI97CER3N0
vr4MkhoXe0rZigAAAABJRU5ErkJggg==" alt="Roter Punkt" />

//CSS
div {
    background: url('
iVBORw0KGgoAAAANSUhEUgAAABAAAAAQAQMAAAAlPW0iAAAABlBMVEUAAAD/
//+l2Z/dAAAAM0lEQVR4nGP4/5/h/1+G/58ZDrAz3D/McH8yw83NDDeNGe4U
g9C9zwz3gVLMDA/A6P9/AFGGFyjOXZtQAAAAAElFTkSuQmCC')
  top left no-repeat;
}

//JavaScript
window.open('data:text/html;charset=utf-8,%3C%21DOCTYPE%20html%3E%0D%0A%3Cht'+
  'ml%20lang%3D%22en%22%3E%0D%0A%3Chead%3E%3Ctitle%3EEmbedded%20Window%3C%2F'+
  'title%3E%3C%2Fhead%3E%0D%0A%3Cbody%3E%3Ch1%3E42%3C%2Fh1%3E%3C%2Fbody%3E%0'+
  'A%3C%2Fhtml%3E%0A%0D%0A','_blank','height=300,width=400');
```

WebSQL

```
var db = window.openDatabase("DBName", "1.0", "description", 5*1024*1024);

db.transaction(function(tx) {
  tx.executeSql("SELECT * FROM todo", [], successCallback, errorCallback);
  tx.executeSql("SELECT * FROM todo WHERE id = ?", [1], successCallback, errorCallback);
  tx.executeSql("INSERT INTO todo (todo) VALUES (?)", [todoItem.content], successCallback, e
  tx.executeSql("UPDATE todo SET todo = ? WHERE id = ?", [todoItem.content, todoItem.id], su
  tx.executeSql("DELETE FROM todo WHERE id = ?", [todoItem.id], successCallback, errorCallba
});
```

Storage

```
//simple String-based Key/Value

//localStorage - per Browser and Domain
//sessionStorage - per Browser-Session and Domain


//set
window.localStorage.setItem('value', JSON.stringify({}));

//get
window.localStorage.getItem('value');

//remove
window.localStorage.removeItem('value');
```

IndexDB

```
var indexedDB      = window.indexedDB || window.webkitIndexedDB || window.mozIndexedDB ||
var IDBTransaction = window.IDBTransaction || window.webkitIDBTransaction;
var IDBKeyRange    = window.IDBKeyRange || window.webkitIDBKeyRange;

var db      = null;
var request = indexedDB.open("todo");
request.onfailure = onError;
request.onsuccess = function(e) {
 db = request.result;

 var v = "1.0";
 // We can only create Object stores in a setVersion transaction;
 if(v != db.version)
 {
  var verRequest       = db.setVersion(v);
  verRequest.onfailure = onError;
  verRequest.onsuccess = function(e) {
   // onsuccess is the only place we can create Object Stores
   var store = db.createObjectStore(
    "todo",
    {
     keyPath: "id", // unique field
     autoIncrement: true
    }
```

File API

```
//FileReader - FileWriter API


window.storageInfo  = window.storageInfo || window.webkitStorageInfo;
window.requestFileSystem  = window.requestFileSystem || window.webkitRequestFileSystem;
window.BlobBuilder = window.BlobBuilder || window.WebKitBlobBuilder;

var onError = function(e) {
 var msg = '';

 switch(e.code)
 {
  case FileError.QUOTA_EXCEEDED_ERR:
   msg = 'QUOTA_EXCEEDED_ERR';
   break;
  case FileError.NOT_FOUND_ERR:
   msg = 'NOT_FOUND_ERR';
   break;
  case FileError.SECURITY_ERR:
   msg = 'SECURITY_ERR';
   break;
  case FileError.INVALID_MODIFICATION_ERR:
   msg = 'INVALID_MODIFICATION_ERR';
   break;
  case FileError.INVALID_STATE_ERR:
   msg = 'INVALID_STATE_ERR';
```

Network Connectivity

```
document.addEventListener('online', function(){

    var isOnLine = navigator.onLine;

});

document.addEventListener('offline', function(){

    var isOnLine = navigator.onLine;

});
```

Storage Limitations

(Quotas)

	App Cache	Web Storage	WebSQL	IndexedDB	File API
iOS 5.1	10 MB	5 MB	5 MB	5 MB	
Android 4	unlimited	5 MB	?	?	
Safari 5.2	unlimited	5 MB	5 MB	5 MB	
Chrome 18	5 MB	5 MB	unlimited	unlimited	unlimited
IE 10	50 MB	10 MB	500 MB	500 MB	
Opera 11	50 MB	5 MB	5 MB	5 MB	
Firefox 11	unlimited	10 MB	50 MB	50 MB	

Editable

```
//Page
document.designMode = 'on';

//Area
<button id="editorBtn" type="button">Enable Editing</button>
<section id="editor" contenteditable="true">


var editorBtn = document.getElementById('editorBtn');
var element = document.getElementById('editor');

editorBtn.addEventListener('click', function(e) {
  e.preventDefault();

  if (element.isContentEditable) {
    // Disable Editing
    element.contentEditable = 'false';
    editorBtn.innerHTML = 'Enable Editing';

    // You could save any changes here.
  } else {
    element.contentEditable = 'true';
    editorBtn.innerHTML = 'Disable Editing';
  }
});
```

Drag'nDrop

